

ME 504: Numerical Methods for Engineering Computation

Course Code	ME 504
Course Name	Numerical Methods for Engineering Computation
L-T-P-C	3-0-0-3
Prerequisites	Consent of the faculty member
Intended for	B.Tech.
Distribution	Elective
Approval	9th Senate (May 2016), in the M.Sc Applied Mathematics curriculum

Course Contents

- **Introduction**

Differential equations in engineering applications. Analytical vs. Numerical solution of a mathematical model. Computer representation of numbers. Errors in numerical computation. Review of CPP programming concepts - Program structure, data types, arrays, structures, functions, file handling using simple problems: Second moment of area, analysis of beams, basic statistics etc. [3 Lectures]

- **Systems of linear algebraic equation**

Gauss elimination, Gauss Jordan, LU decomposition, and Gauss-Seidel methods. Thomas algorithm for tri-diagonal and Cholesky decomposition for symmetric matrices. Matrix inversion methods. CPP programs for the mentioned schemes. [6 Lectures]

- **Eigen problem**

Eigenvalues and Eigen vectors. Properties. Methods of estimation of Eigenvalues and Eigenvectors – Power Iteration Methods, Jacobi Iteration, QR algorithm; Application of these concepts towards Matrix Inversion and solution of linear simultaneous equations. [5 Lectures]

- **Curve fitting**

Linear regression, polynomial regression, nonlinear regression. CPP program for constitution of normal equations [4 Lectures]

- **Finite differences and interpolation**

Taylor's series, Forward, backward and central differences, Difference tables, Finite difference operators, Newton's forward and backward interpolation formulae, Stirling's, Bessel's and Laplace-Everett's interpolation formulae. Lagrange's polynomial and Newton's divided difference formula. CPP programs for implementing interpolation schemes. [6 Lectures]

- **Numerical differentiation and integration**

Derivatives using forward, backward and central difference formulae. Newton-Cotes integration formulae – Trapezoidal rule, Simpson's 1/3 rule, Simpson's 3/8 rule. CPP programs for implementing numerical integration schemes. [6 Lectures]

- **ODEs**

Description of initial and boundary value problems. Taylor's series, Picard's, Euler's, Runge-Kutta and Milne's methods. Finite difference solution of boundary value problems. [6 Lectures]

- **PDEs**

Description of Elliptic, Parabolic and Hyperbolic PDEs. Solution of Laplace's and Poisson's equations. Solution of linear heat transfer equation - CPP programs for implementation of explicit and implicit schemes. Conditions for stability. [6 Lectures]

Text Books

1. Steven C. Chapra and Raymond P. Canale, **Numerical methods for engineers**, McGraw- Hill. 2015.
2. Nita H. Shah, **Numerical methods with C++ programming**, PHI. 2009.

References

1. Joe D. Hoffman, **Numerical methods for engineers and scientists**, Marcel Dekker Inc., 2001.
2. T. Veerarajan and T. Ramachandran, **Numerical methods with programs in C**, Tata McGraw-Hill, 2006.